

02464 Artificial Intelligence and Human Cognition

Overview

Teachers



Ivana

Tobias



Teaching assistant week 1-2



Kathrine

Motivation

- AI systems inspire models of human cognition
 - Perception and action as Bayesian inference
- How can we compare human and computers?
 - Human intelligence vs artificial intelligence
 - Learning
 - Memory
- Can we think of AI as a reverse engineering problem?
 - Not generally
 - Neural networks, language models and automatic speech recognition
- AI systems learn from humans
 - Mimicking human behavior
 - Image classification (typically human labels, not ground truth)
- AI systems are evaluated by humans
 - Language - who is the judge of how well GPT works?

General info

- Go to 'overview' under content on DTU Learn for general course info

Exam

- The final written exam is on 15 December 2025. We do not know the time or place for the exam yet. We will announce it when we know.
- Your grade in the course will be based 80% on the final exam, and 20% on the submitted exercises (10% per exercise).

Reports / exercises

- The exercises are a key part of the course curriculum and will be part of the exam. Completing the exercises and lectures will prepare you for the exam questions.
- There are two mandatory exercises in the course
 - (1 for Tobias' half, 1 for Ivana's)

Course structure

See 'overview' under 'content' on DTU Learn

Topic	Teacher	Teaching week
Introduction to cognitive science	Tobias	1 & 6
Perception as Bayesian Inference	Tobias	2-3
Neural coding of visual information	Tobias	4-5
Motor control	Ivana	7
Social cognition	Ivana	8-9
Embodied cognition and psychophysiology	Ivana	10-12

Intro to Cognitive Science - week 1 & 6

How can we compare humans and computers?

- Intelligence
 - How do we define intelligence?
- Computation
 - Marr's three levels
- Learning
 - Machine learning paradigms as models of human learning
- Memory
 - How do we measure human memory capacity?

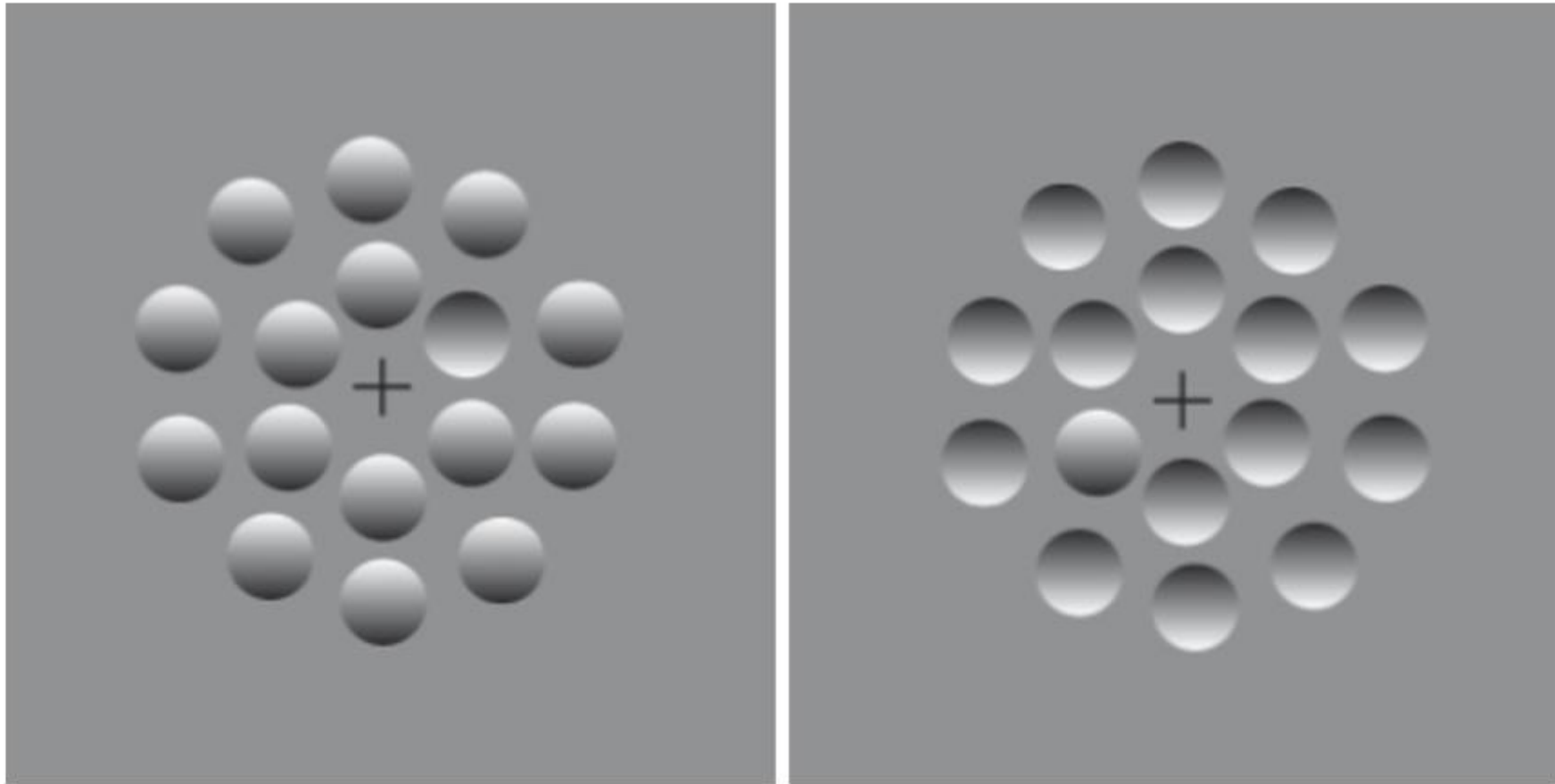
Perception/Bayes – week 2-3

What's the fuzz?!

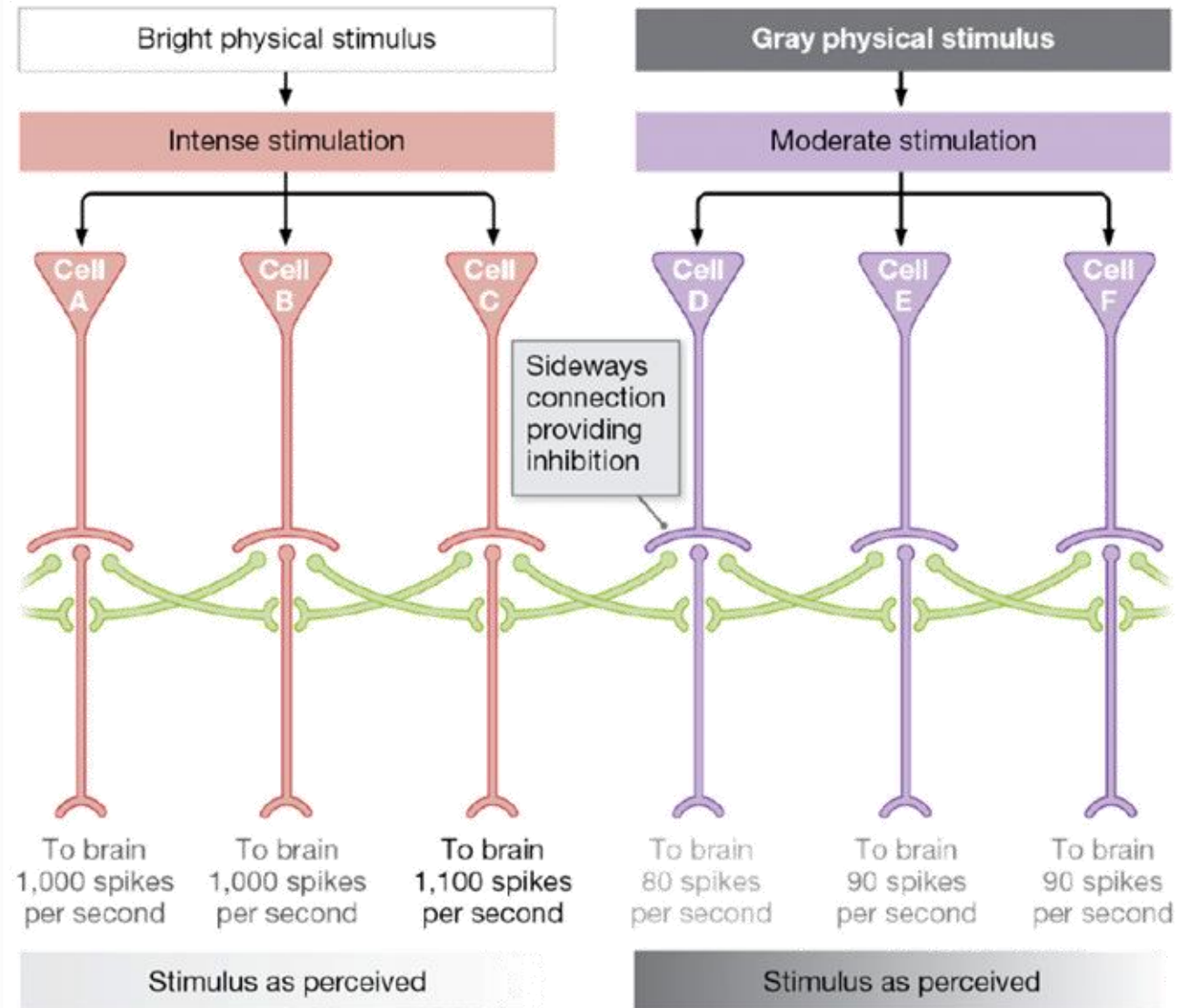
$$P(w \mid d) = \frac{P(d \mid w)P(w)}{P(d)}$$

- The problem of *any* observer
- The problem of perception
 - Probability of the world given sensory *data*
- Likelihood contains a model of the world
 - Not enough information
 - Prior adds information

Perception/Bayes – week 2-3



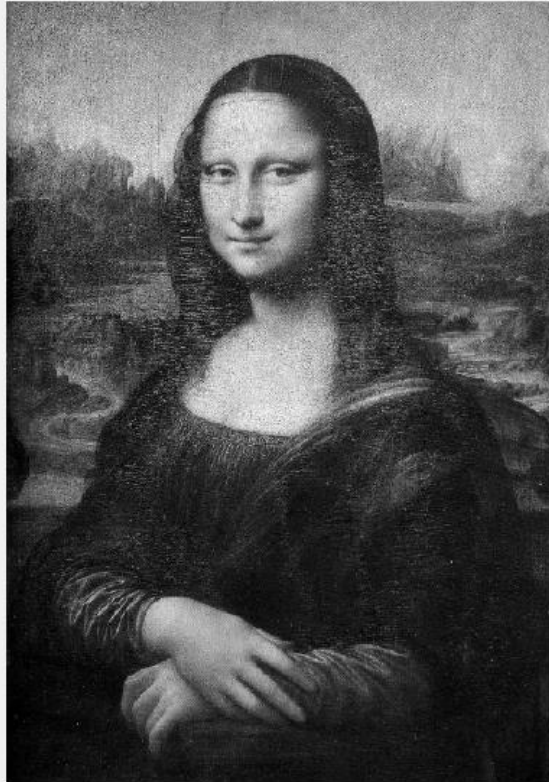
Neural coding – week 4-5



Neural coding – week 4-5

Mona Lisa

Original



Filtered

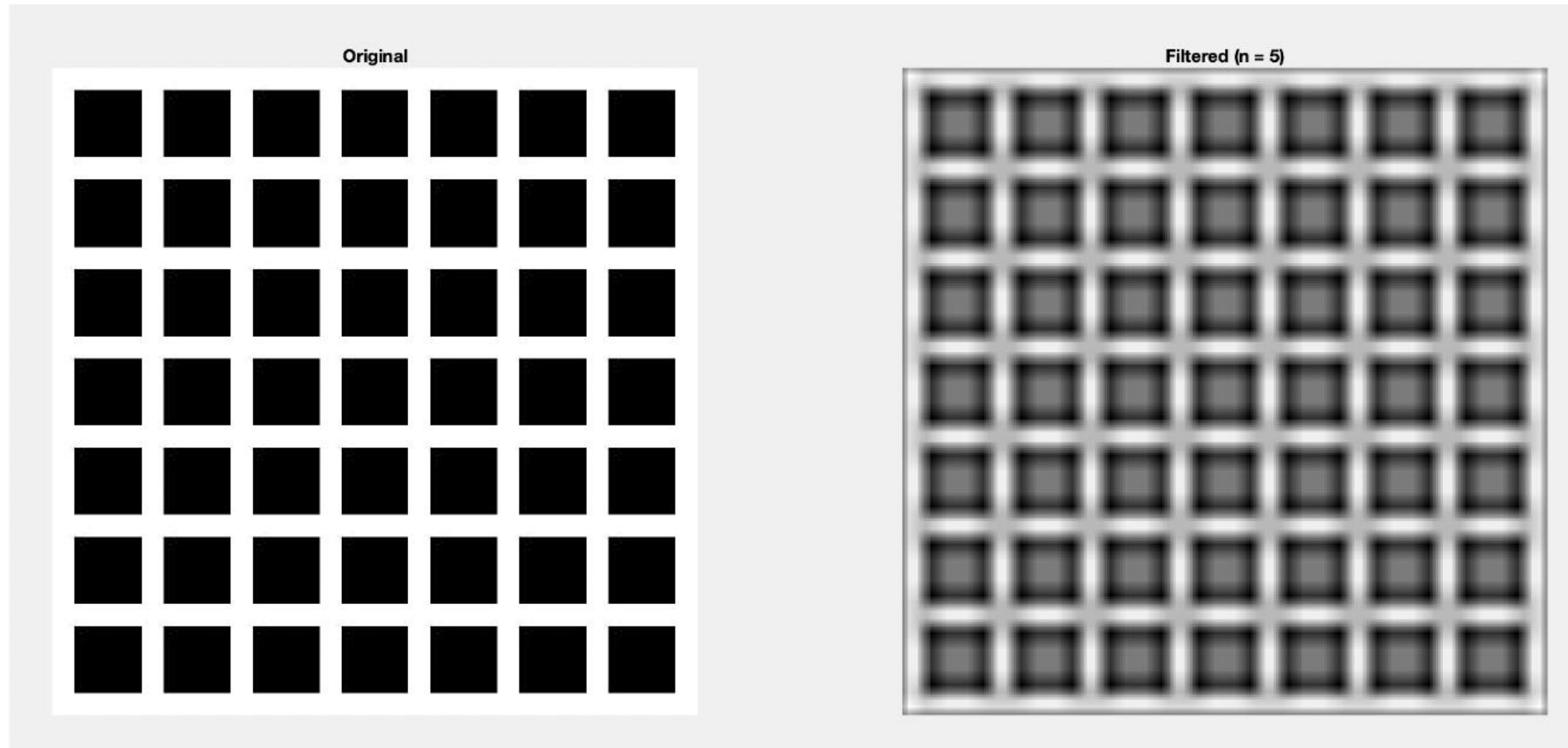


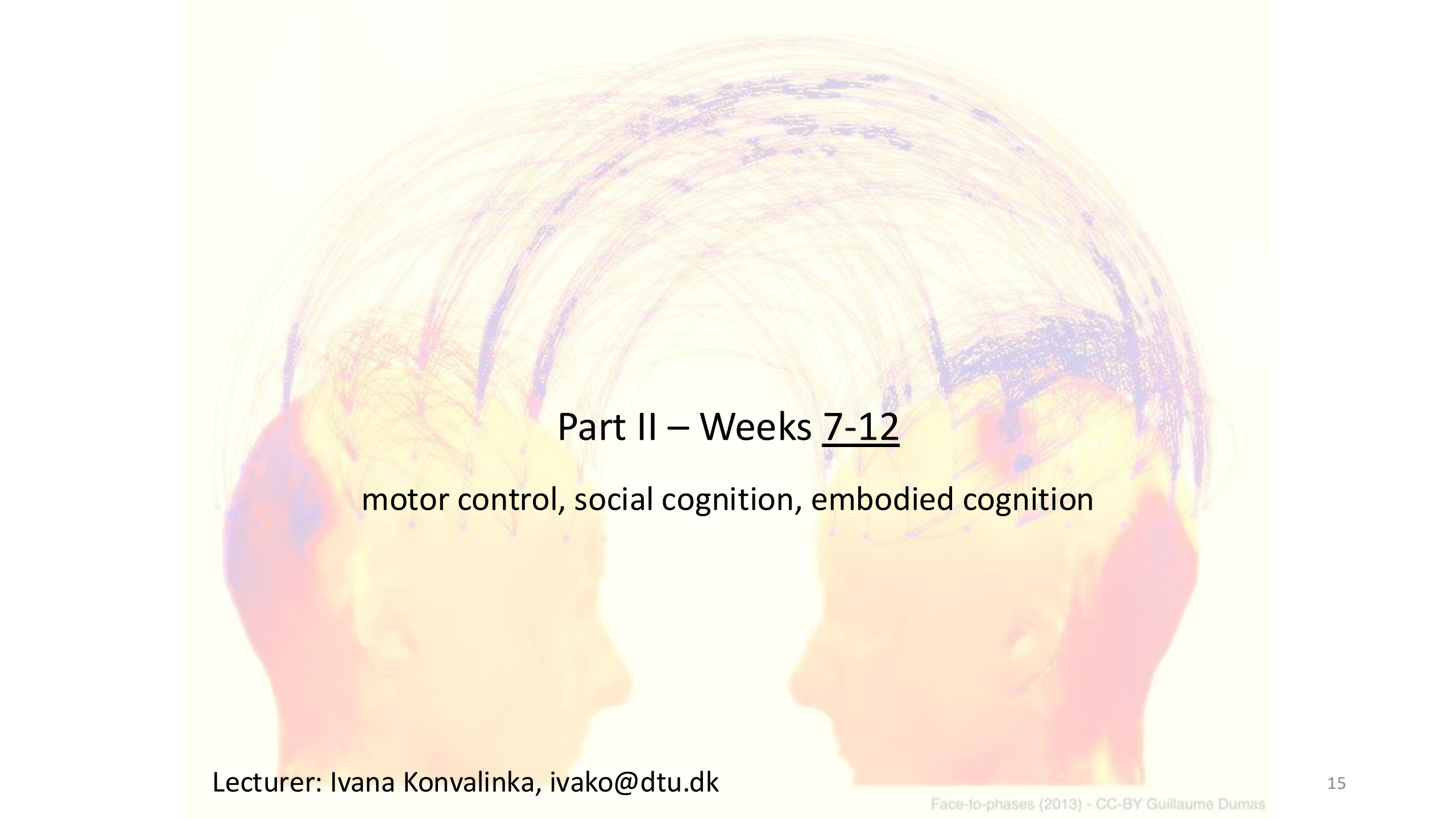
Thresholded



Neural coding – week 4-5

The Hermann Grid

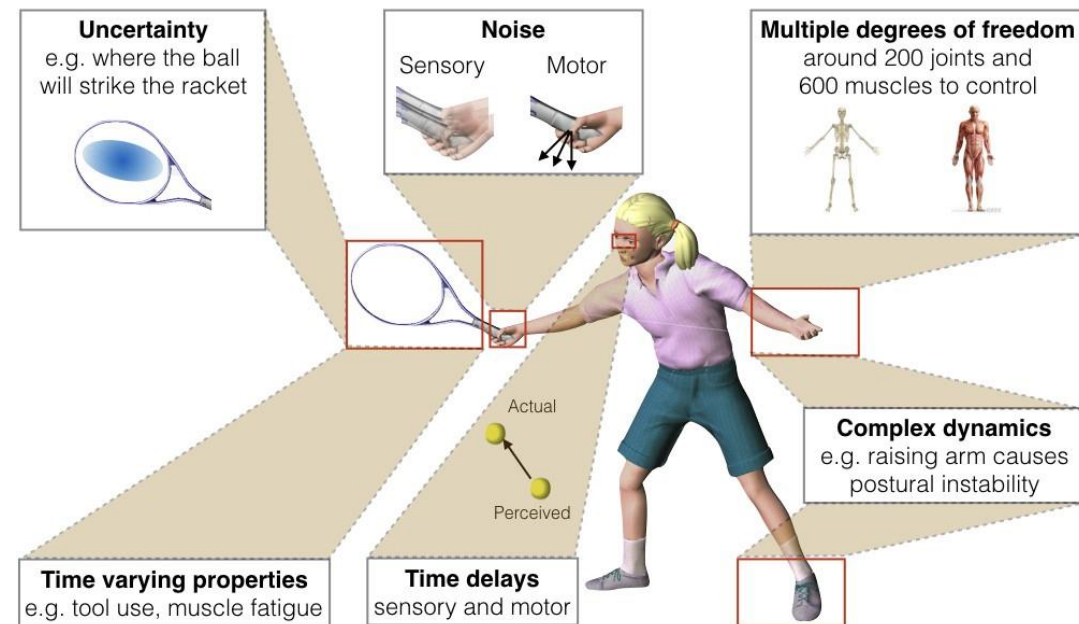
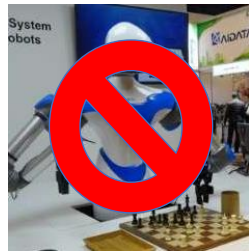




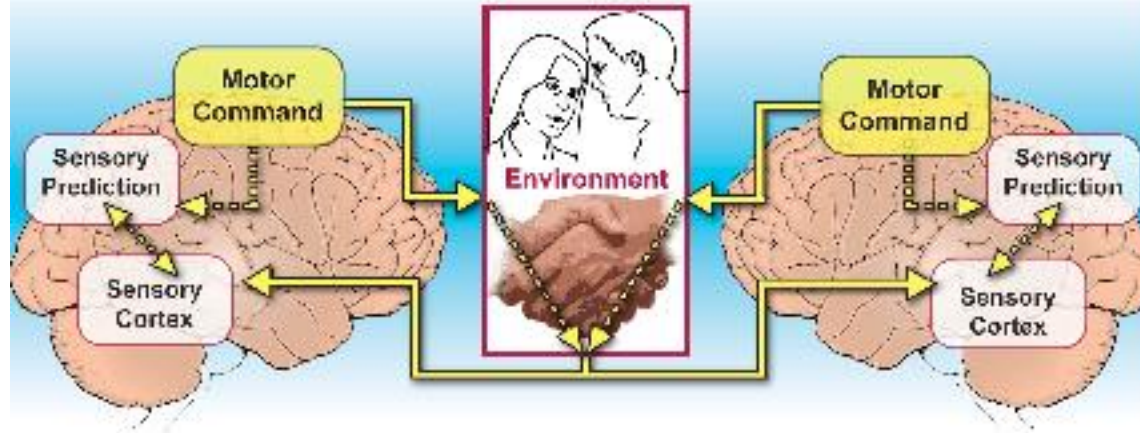
Part II – Weeks 7-12

motor control, social cognition, embodied cognition

Motor Control



Social Cognition



- your action output = another person's perceptual input (and vice-versa)
- what are the cognitive and neural mechanisms that enable us to successfully interact with other people?
- using AI to understand (social) cognitive mechanisms

Embodied Cognition

- cognition is shaped by aspects of the entire physical body of an agent: it is not restricted to the brain
- example:
 - infer cognitive processes from bodily signals
 - analyze heart rate variability (HRV) as a measure of arousal

